

Guillermo Fernando Farfán Romero

Senior Python Backend Engineer & Technical Lead

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PROFESSIONAL SUMMARY

Senior Python Backend Engineer and Technical Lead with 8+ years of experience designing scalable microservices architectures, high-throughput RESTful APIs, and enterprise integration backends. Specialist in **Python 3.10+ (FastAPI, Asyncio, uvloop, Celery, SQLAlchemy 2.0, Pydantic v2)**, optimized data persistence in **PostgreSQL (with pgvector for semantic search and AI/LLM integrations)** and **Redis**, and containerized cloud workloads on **Google Cloud Platform (Cloud Run, GKE)**.

Strong track record leading the adoption of Clean Architecture, Domain-Driven Design (DDD), automated testing suites with Pytest under TDD methodologies, and distributed observability with **OpenTelemetry / IBM Instana**. Bridges business objectives and technical scalability with strategic team leadership, backed by a dual background in Information Systems and Labor Relations.

TECHNICAL SKILLS & COMPETENCIES

- Languages & Backend Core:** Python 3.10+ (Asyncio, Strict Type Hints, Concurrency, Generators, uvloop), TypeScript / Node.js, Bash Scripting.
- Python Frameworks & Ecosystem:** FastAPI, Pydantic v2, SQLAlchemy 2.0 (Async Engine & ORM), GenAI / LLM Integration (Vertex AI, OpenAI API, RAG, pgvector), Alembic (Zero-Downtime Migrations), Celery, Django / DRF, Flask, httpx, Uvicorn, Gunicorn, Streaming APIs (SSE / WebSockets).
- Databases, Caching & Messaging:** PostgreSQL (Advanced Modeling, pgvector, Composite Indexes, JSONB, asyncpg, PgBouncer), Redis (Distributed Caching, Rate Limiting, Pub-Sub, Celery Broker), MySQL, MongoDB, Google Cloud Pub/Sub.
- Cloud & DevOps:** Google Cloud Platform (Cloud Run, GKE, Cloud Storage, Secret Manager), Docker (Multi-stage builds), Docker Compose, Terraform, GitHub Actions CI/CD.
- Architecture & Engineering Quality:** Microservices, Clean Architecture, Hexagonal Architecture (Ports & Adapters), Domain-Driven Design (DDD), Event-Driven Architecture, Architecture Decision Records (ADRs), REST APIs, OpenAPI/Swagger, TDD/BDD (Pytest, Factory Boy).
- Observability & Reliability:** OpenTelemetry (OTel), IBM Instana APM, Sentry, Prometheus, Git, Postman, SonarQube.

PROFESSIONAL EXPERIENCE

Technical Lead & Senior Systems Engineer (Python Backend Focus) | Reforest Latam

May 2024 – Present | Full-time (Remote)

- Led the adoption of Clean Architecture and Domain-Driven Design (DDD)** across an engineering team of 6 for environmental telemetry platforms, formalizing Architecture Decision Records (ADRs) and **accelerating modular feature delivery by 35%**.
- Established >85% automated test coverage across microservices** by architecting robust test suites with **Pytest** under TDD practices.
- Orchestrated distributed asynchronous workers with Celery and Redis**, integrated with asynchronous **FastAPI** endpoints and **Cloud Pub/Sub** for decoupled, high-scale background event processing.
- Engineered cloud-native security and integration layers** connecting microservices to GCP resources (Cloud Storage, Secret Manager) under JWT/OAuth2 authentication and least-privilege principles.
- Implemented automated CI/CD workflows in GitHub Actions**, enabling continuous **zero-downtime deployments** to **Cloud Run** and **GKE**.

Senior Python & Software Consultant | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- Eliminated N+1 query bottlenecks and optimized asynchronous persistence with SQLAlchemy 2.0** (selective *eager loading*), strict modeling with **Pydantic v2**, and caching with **Redis**, driving a **>60% reduction in p95 latency on critical endpoints**.
- Architected and built ingestion microservices and transactional sync engines** connecting enterprise ERP/SAP platforms with relational databases for international retail, mining, and healthcare corporations.
- Designed high-concurrency, mission-critical REST APIs in FastAPI**, guaranteeing OpenAPI/Swagger contracts, rigorous schema validation, asynchronous streaming, and structured exception handling.
- Standardized reproducible development and production environments** through multi-stage Docker builds and automated infrastructure provisioning with **Terraform** on Google Cloud.

Senior Software Engineer & Database Administrator | University of Buenos Aires (UBA)

March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- Engineered concurrent asynchronous Python microservices and ETL data synchronization pipelines** (+50,000 users), **slashing batch processing runtimes from hours to minutes**.
- Implemented database schema versioning and governance with Alembic**, guaranteeing transactional data integrity and automated testing suites on PostgreSQL and SQL Server.
- Optimized complex SQL queries and stored procedures**, removing computation bottlenecks across high-demand transactional database servers.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)

January 2020 – December 2023 | Full-time (Buenos Aires, Argentina)

- **Established technical interoperability standards and automated validation layers in backend services**, achieving a **40% reduction in cross-departmental data inconsistencies**.
- **Maintained a 99.9% operational uptime SLA** across municipal platforms through code reviews and strict software release governance.

EDUCATION

- **Associate Degree in Information Systems** | National Technological University (UTN) (*In Progress — Expected Graduation: December 2026*)
- **Bachelor's Degree in Labor Relations & Human Resources** | University of Buenos Aires (UBA) (*Graduated — Focus on Organizational Leadership, Talent Strategy & High-Performance Team Dynamics*)

KEY CERTIFICATIONS (100% VERIFIABLE ON CREDLY)

- **Backend & Java Certifications (Credly):** Meta Back-End Developer Certificate | IBM Java Developer Professional Certificate V2 | IBM JavaScript Backend Developer Professional Certificate V2.
- **DevOps & Software Engineering Certifications (Credly):** IBM DevOps and Software Engineering Professional Certificate | IBM Full Stack Software Developer Certificate | Meta Full-Stack Engineer Certificate.

LANGUAGES

- **Spanish:** Native
- **English:** Full Professional Proficiency (C1 / Full technical fluency for backend architecture, live engineering syncs, RFCs, and technical documentation)